

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA5116 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5

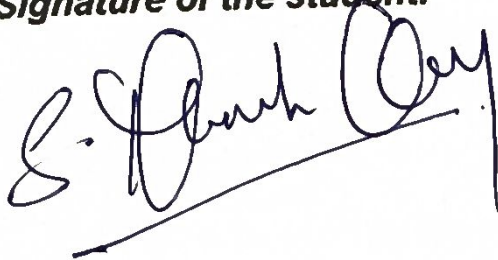
TOTAL MARKS: 50

Annexure A
MCQ

Code of Conduct:

I Dhanush Kumar .S ¹⁹²⁴²¹⁰⁹⁶ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A
MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash?
A) Avalanche effect
☒ B) Determinism
C) Collision resistance
D) Diffusion
2. What is the primary purpose of SHA-3?
A) Faster encryption
☒ B) Alternative secure hashing standard
C) Data compression
D) Key exchange
3. Which attack tries all possible inputs to match a hash?
A) Replay attack
☒ B) Brute-force attack
C) MITM attack
D) DoS attack
4. What is a salted hash?
A) Encrypted hash
☒ B) Hash with random data added
C) Shortened hash
D) Reversed hash
5. Which algorithm is widely used for password hashing?
A) AES
B) RSA
☒ C) bcrypt
D) DES
6. What does second pre-image resistance ensure?
A) Same input produces same output
☒ B) Hard to find different input with same hash
C) Fast hashing
D) Small hash size
7. What is the output size of SHA-512?
A) 256 bits
B) 128 bits
☒ C) 512 bits
D) 1024 bits
8. Which technique slows down brute-force attacks?
A) Encryption
B) Key exchange
☒ C) Key stretching
D) Compression
9. What is the purpose of a nonce?
A) Encryption
☒ B) Prevent replay attacks
C) Compress data
D) Generate keys
10. Which function is irreversible?
A) Encryption
☒ B) Hashing

- C) Compression
- D) Encoding
- 11. Which algorithm is used in DSS?
 - A) AES
 - ☒ B) DSA
 - C) DES
 - D) RC4
- 12. What is required to verify a digital signature?
 - A) Private key
 - ☒ B) Public key
 - C) Session key
 - D) Shared key
- 13. What does a digital signature guarantee?
 - A) Confidentiality
 - ☒ B) Integrity and authenticity
 - C) Compression
 - D) Speed
- 14. What happens if message digest changes?
 - A) Signature remains valid
 - ☒ B) Verification fails
 - C) Encryption fails
 - D) Speed increases
- 15. Which attack involves forging a signature?
 - A) Replay
 - ☒ B) Forgery
 - C) DoS
 - D) Sniffing
- 16. What is the role of a hash in signatures?
 - A) Encrypt data
 - B) Reduce size
 - ☒ C) Ensure integrity
 - D) Store keys
- 17. Which key is kept secret in signatures?
 - A) Public key
 - ☒ B) Private key
 - C) Session key
 - D) Shared key
- 18. What ensures trust in certificates?
 - A) Encryption
 - ☒ B) Certificate Authority
 - C) Compression
 - D) Hash size
- 19. What is certificate revocation?
 - A) Key generation
 - ☒ B) Canceling validity
 - C) Encryption
 - D) Compression
- 20. What is OCSP used for?
 - A) Encryption
 - ☒ B) Certificate validation
 - C) Compression
 - D) Routing
- 21. What is the main goal of Kerberos tickets?
 - A) Encryption
 - ☒ B) Authentication

- C) Compression
- D) Routing
- 22. Which protocol uses tickets for authentication?
 - A) SSL
 - ☒ B) Kerberos
 - C) FTP
 - D) HTTP
- 23. What is a session key?
 - A) Permanent key
 - ☒ B) Temporary key
 - C) Public key
 - D) Private key
- 24. What is mutual authentication?
 - A) One-way authentication
 - ☒ B) Both parties verify each other
 - C) Anonymous login
 - D) Weak authentication
- 25. What ensures freshness in Kerberos?
 - A) Encryption
 - ☒ B) Timestamp
 - C) Compression
 - D) Hashing
- 26. What is a replay attack?
 - A) Data modification
 - ☒ B) Resending valid data
 - C) Encryption
 - D) Compression
- 27. What is the function of an authenticator in Kerberos?
 - A) Data storage
 - ☒ B) Identity proof
 - C) Encryption
 - D) Compression
- 28. What happens if system clocks differ in Kerberos?
 - A) Faster authentication
 - ☒ B) Failure
 - C) Better security
 - D) No effect
- 29. What is the KDC composed of?
 - A) Client & Server
 - ☒ B) AS & TGS
 - C) Firewall & IDS
 - D) Router & Switch
- 30. What is the AS in Kerberos?
 - A) Application Server
 - ☒ B) Authentication Server
 - C) Access Server
 - D) Admin Server
- 31. What is X.509 certificate format used in?
 - A) Networking
 - ☒ B) Digital certificates
 - C) Compression
 - D) Routing
- 32. What identifies the certificate owner?
 - A) Public key
 - ☒ B) Subject field

- C) Signature
 - D) Hash
33. What ensures certificate integrity?
- A) Encryption
 - ☒ B) Digital signature
 - C) Compression
 - D) Routing
34. What is a root certificate?
- A) Expired certificate
 - ☒ B) Self-signed trusted certificate
 - C) Invalid certificate
 - D) Temporary certificate
35. What is certificate pinning?
- A) Storing keys
 - ☒ B) Binding certificate to host
 - C) Encryption
 - D) Compression
36. What is the risk of weak CA validation?
- A) Faster speed
 - ☒ B) MITM attacks
 - C) Compression
 - D) Routing
37. What is the purpose of trust chain?
- A) Encryption
 - ☒ B) Validate certificate hierarchy
 - C) Compress data
 - D) Routing
38. What is digital certificate used for?
- A) Compression
 - ☒ B) Authentication
 - C) Routing
 - D) Storage
39. What ensures non-repudiation?
- A) Encryption
 - ☒ B) Digital signature
 - C) Compression
 - D) Routing
40. What is a hash collision?
- A) Same input
 - ☒ B) Same output for different inputs
 - C) Encryption
 - D) Compression
41. Which hash is strongest?
- A) MD5
 - B) SHA-1
 - ☒ C) SHA-256
 - D) CRC
42. What is a one-way function?
- A) Reversible
 - ☒ B) Irreversible
 - C) Fast
 - D) Small
43. What ensures authentication in PKI?
- A) Encryption
 - ☒ B) Certificates

- C) Compression
- D) Routing
- 44. What is key escrow?
 - A) Key deletion
 - ☒ B) Key storage by third party
 - C) Encryption
 - D) Compression
- 45. What is identity spoofing?
 - A) Encryption
 - ☒ B) Fake identity
 - C) Compression
 - D) Routing
- 46. What is trust anchor?
 - ☒ A) Root CA
 - B) Client
 - C) Server
 - D) Router
- 47. What ensures integrity of message?
 - A) Encryption
 - ☒ B) Hash
 - C) Compression
 - D) Routing
- 48. What is digital envelope?
 - A) Hash
 - ☒ B) Encrypted message + key
 - C) Compression
 - D) Routing
- 49. What is certificate fingerprint?
 - A) Encryption
 - ☒ B) Hash of certificate
 - C) Compression
 - D) Routing
- 50. What is PKI used for?
 - A) Routing
 - ☒ B) Secure communication
 - C) Compression
 - D) Storage